

Jawaharlal Nehru Govt. Engineering
College Sundernagar, Distt. Mandi (H.P.) -175018
Phone No. 01907-267199, 267688, Fax No. 267504
Web site: www.jngec.ac.in, Email: jngechp@yahoo.co.in

2nd Periodical Examination (Nov, 2024)

Course Name: - Digital System Design Course Code :-(ECEPC-312) Schemes: NEP

Branch:- CSE (AI&ML) Semester:-3rd Time:-2 Hours

Max Marks: -50

Note:-Attempt all questions.

Course Outcomes:

CO1: To demonstrate understanding of Boolean algebra and binary arithmetic operations.

CO2: To identify and describe MSI and their applications in digital circuits.

CO3: To explain the operation and design considerations for synchronous circuits.

CO4: To interpret and differentiate between dataflow, behavioral, and structural modeling in VHDL.

Q1. (a) CO2 What is the primary advantage of TTL technology?

(b) CO2 Why is ECL considered one of the fastest logic families?

(c) CO3 What is a key benefit of CMOS technology?

(d) CO3 How do flip-flops function in digital circuits?

(e) CO3 Define use of flip flop.

(5*2)

Q2. Discuss drawback of TTL and how it can be overcome in Totem Pole configuration, explain with neat diagram.

CO4, 10

Q3. Implement Full adder and Full subtractor by using PROM 8K*4.

CO4, 10

Q4. Discuss Race around condition, explain with neat diagram how it can be removed.

CO3, 10

Q5. Design 3-bit synchronous counter by using JK flip flop.

CO3, 10

Course advisor: Rita Devi